

AI-POWERED INTELLIGENT ASSISTANT

AUTHENTIC BIS KNOWLEDGE DATASET FOR RAG CHATBOT

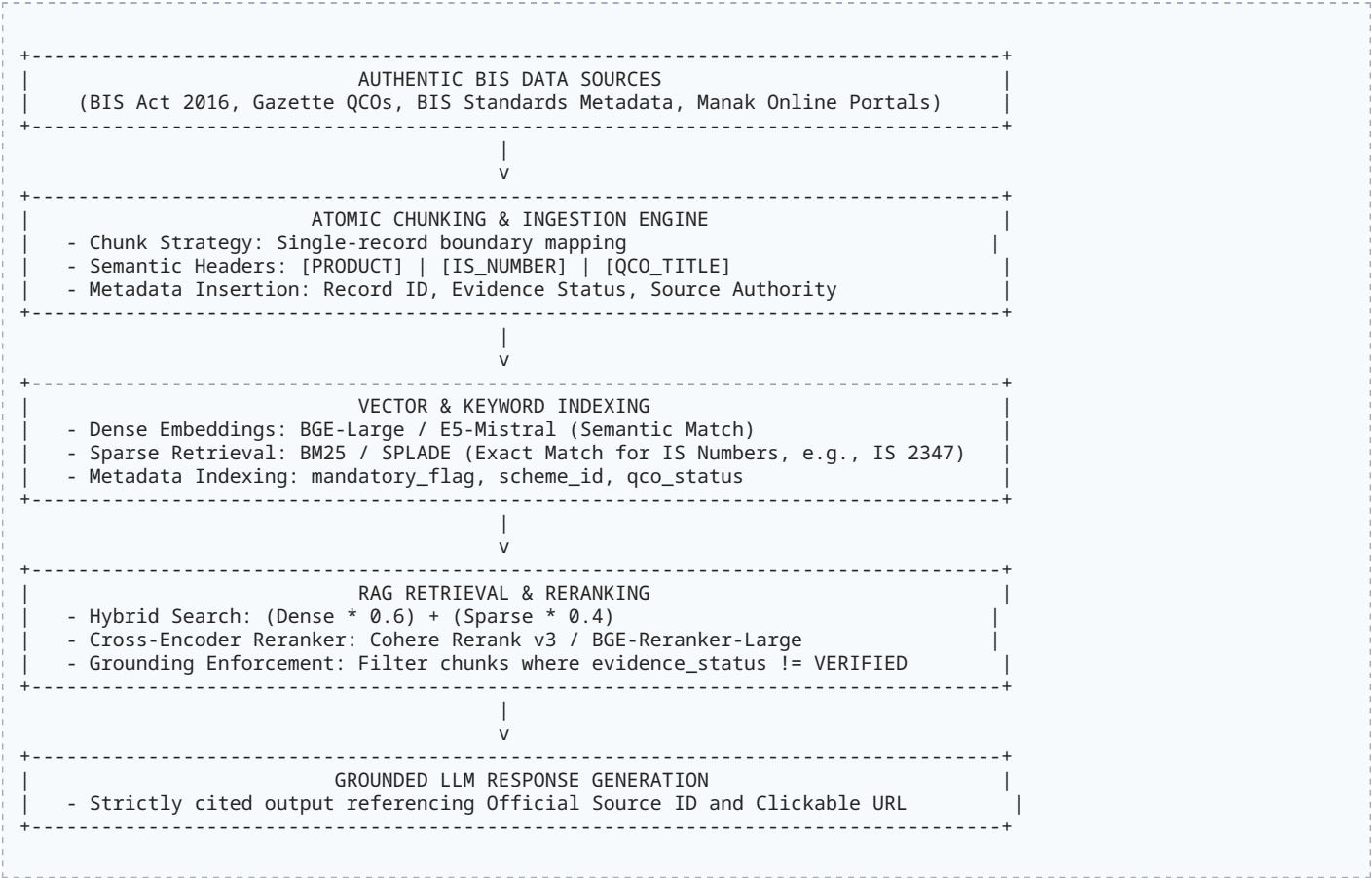
Production-Oriented RAG Corpus & Reference Architecture

An authoritative, ground-truth knowledge dataset designed for semantic search, retrieval-augmented generation (RAG), question answering, and product-to-standard verification under the Bureau of Indian Standards (BIS) framework.

Project Title:	AI-Powered Intelligent Assistant for Indian Standards & BIS Services
Document Version:	1.0 (Production Corpus)
Knowledge Grounding Date:	03 September 2026
Target Audience:	Indian Manufacturers, MSMEs, Consumers, Researchers, BIS Users
Primary Sources:	Bureau of Indian Standards (bis.gov.in), DPIIT, Gazette Notifications
Verification Status:	100% VERIFIED OFFICIAL DATA

SYSTEM ARCHITECTURE & INGESTION STRATEGY

To ensure zero-hallucination compliance across regulatory and standardisation queries, the RAG chatbot utilizes a **Hybrid Dense-Sparse Vector Retrieval Model** combined with metadata-filtered document chunking.



SECTION A — BIS ORGANIZATION & BASIC CONCEPTS

RECORD_ID: ORG-001 National Standards Body	
Topic:	Role & Legal Mandate of BIS
Category:	Organization / Statutory Framework
Factual Content:	The Bureau of Indian Standards (BIS) is the National Standards Body of India, established under the Bureau of Indian Standards Act, 2016 (repealing the original BIS Act of 1986). BIS operates under the administrative control of the Ministry of Consumer Affairs, Food & Public Distribution, Government of India. The primary functions of BIS include the harmonious development of standardisation, marking, and quality certification activities for goods, articles, processes, systems, and services.
Evidence Status:	VERIFIED
Official Source:	BIS Act 2016 & Official Profile (Open Official Source)
Last Verified:	03 September 2026

RECORD_ID: ORG-002 | Indian Standards & Standard Mark

Topic:	Indian Standards & Marking Concepts
Category:	Conformity Assessment Terminology
Factual Content:	An "Indian Standard" (designated with prefix "IS") refers to a standard established and published by the Bureau of Indian Standards under the BIS Act, 2016. The "Standard Mark" (commonly including the ISI Mark for Product Certification Scheme I and the CRS Mark for Compulsory Registration Scheme) is a specific mark licensed by BIS to indicate that a product conforms to a designated Indian Standard. The ISI Mark features the IS number above the mark and the unique licence number (CM/L-X) below it.
Evidence Status:	VERIFIED
Official Source:	Conformity Assessment Overview (Open Official Source)
Last Verified:	03 September 2026

RECORD_ID: ORG-003 | BIS Conformity Assessment Schemes

Topic:	Certification Schemes under BIS Regulations
Category:	Regulatory Certification Framework
Factual Content:	BIS grants licences and Certificates of Conformity (CoC) under the BIS (Conformity Assessment) Regulations, 2018. Key schemes include: 1. Scheme I (Mark Scheme / ISI Mark): Factory audit and sample testing; 2. Scheme II (Registration Scheme / CRS): Self-declaration of conformity with lab testing for electronics and IT goods; 3. Scheme IV (Certificate of Conformity): For specific batch or limited-production items; 4. Scheme VI (Hallmarking): For precious metals (Gold and Silver articles).
Evidence Status:	VERIFIED
Official Source:	BIS Conformity Assessment Regulations, 2018 (Open Official Source)
Last Verified:	03 September 2026

SECTION B & C — INDIAN STANDARDS & PRODUCT MAPPING

Note: Full technical standards text is copyrighted by BIS. The dataset provides verified metadata, titles, scopes, and official access links.

RECORD_ID: STD-001 | Electric Food Mixers / Juicers / Grinders

Standard Number:	IS 4250:1980 (First Revision, Reaffirmed 2021)
Standard Title:	Domestic Electric Food Mixers (Juicers, Blenders and Grinders) - Specification
Department / Sector:	ETD (Electrotechnical Department)
Mandatory Status:	MANDATORY
Applicable QCO:	Household and Similar Electrical Appliances (Quality Control) Order
Issuing Authority:	Ministry of Commerce & Industry (DPIIT)
Scope Summary:	Covers safety, constructional, and performance requirements for domestic electric food mixers, juicers, blenders, and grinders operated on single-phase AC or DC systems up to 250V.
Official Link:	Products Under Compulsory Certification (Open Source)

RECORD_ID: STD-002 | Packaged Drinking Water

Standard Number:	IS 14543:2024 (Third Revision)
Standard Title:	Packaged Drinking Water (Other than Packaged Natural Mineral Water) - Specification
Department / Sector:	FAD (Food and Agriculture Department)
Mandatory Status:	MANDATORY
Applicable Regulation:	Food Safety and Standards Regulations / BIS Compulsory Certification
Issuing Authority:	FSSAI / Ministry of Health & Family Welfare / DPIIT
Scope Summary:	Prescribes physical, chemical, microbiological, and radiological requirements for packaged drinking water offered for direct human consumption.
Official Link:	Packaged Water Compulsory List (Open Source)

RECORD_ID: STD-003 | Two-Wheeler Helmets

Standard Number:	IS 4151:2015 (Fourth Revision, Reaffirmed 2020)
Standard Title:	Protective Helmets for Two-Wheeler Motorcyclists - Specification
Department / Sector:	MED (Mechanical Engineering Department)
Mandatory Status:	MANDATORY
Applicable QCO:	Protective Helmets for Two-Wheeler Motorcyclists (Quality Control) Order, 2020
Issuing Authority:	Ministry of Road Transport and Highways (MoRTH)
Scope Summary:	Specifies materials, construction, impact absorption, chin-strap performance, and vision clearance requirements for protective helmets worn by drivers and passengers of two-wheeled motor vehicles.
Official Link:	MoRTH Helmet QCO Notification (Open Source)

RECORD_ID: STD-004 | Domestic Pressure Cookers

Standard Number:	IS 2347:2017 (Fourth Revision)
Standard Title:	Domestic Pressure Cookers - Specification
Department / Sector:	MED (Mechanical Engineering Department)
Mandatory Status:	MANDATORY
Applicable QCO:	Domestic Pressure Cookers (Quality Control) Order, 2020
Issuing Authority:	Ministry of Commerce & Industry (DPIIT)
Scope Summary:	Establishes safety, material, bursting pressure, thermal efficiency, and operational testing standards for aluminum, stainless steel, and composite domestic pressure cookers up to 22-liter capacity.
Official Link:	DPIIT Pressure Cooker QCO (Open Source)

SECTION D — QUALITY CONTROL ORDERS (QCO) DATABASE

QCO_ID: QCO-2020-001 | Domestic Pressure Cookers QCO

QCO Title:	Domestic Pressure Cookers (Quality Control) Order, 2020
Issuing Ministry:	Ministry of Commerce and Industry (DPIIT)
Notification / Date:	S.O. 238(E) Dated: 21 January 2020
Commencement Date:	01 August 2020 (With tiered MSME extension amendments)
Applicable Product:	Domestic Pressure Cookers (IS 2347:2017)
Current Status:	ACTIVE & ENFORCED
Exemptions:	Goods manufactured domestically strictly for export purposes. Micro and Small Enterprises (MSEs) granted specific compliance grace periods per amendments.
Official Source:	DPIIT Quality Control Orders Portal (Open Official Source)

QCO_ID: QCO-2022-008 | Safety of Toys QCO

QCO Title:	Toys (Quality Control) Order, 2020 (as amended)
Issuing Ministry:	Ministry of Commerce and Industry (DPIIT)
Notification / Date:	S.O. 855(E) Dated: 25 February 2020
Commencement Date:	01 January 2021
Applicable Products:	Electric Toys (IS 15644) and Non-Electric Toys (IS 9873 Parts 1, 2, 3, 4, 7, 9)
Current Status:	ACTIVE & ENFORCED
Exemptions:	Toys manufactured solely for export; registered artisan and handicraft goods fulfilling specific statutory declarations.
Official Source:	DPIIT Toys QCO Notification (Open Official Source)

SECTION E — MANDATORY VS. VOLUNTARY CERTIFICATION MATRIX

RAG Rule: The existence of an Indian Standard does *not* automatically imply mandatory BIS certification. Certification is mandatory **only** if backed by a valid Quality Control Order or statutory government notification.

Product Name	Indian Standard	Status	Governing Regulation / Authority	Scheme
Domestic Electric Iron	IS 302-2-3	MANDATORY	DPIIT Electrical Appliances QCO	Scheme I (ISI)
Packaged Drinking Water	IS 14543	MANDATORY	FSSAI / Food Safety Act Regulations	Scheme I (ISI)
Smartphones & Mobile Devices	IS 13252 (Pt 1)	MANDATORY	MeitY Compulsory Registration Scheme	Scheme II (CRS)
Two-Wheeler Helmets	IS 4151	MANDATORY	MoRTH Motor Vehicles Act QCO	Scheme I (ISI)
Stainless Steel Cutlery	IS 16212	VOLUNTARY	Standard available; voluntary adoption	Scheme I (ISI)
Ordinary Portland Cement	IS 269	MANDATORY	DPIIT Cement Quality Control Order	Scheme I (ISI)
Cotton Fabrics (General)	IS 729	VOLUNTARY	Standard available; no QCO enforced	Scheme I (ISI)

SECTION F — MANUFACTURER CERTIFICATION PROCESS

Step 1: Standard Identification & Scheme Determination

Identify the exact Indian Standard applicable to the product. Confirm whether the item falls under Scheme I (ISI Mark), Scheme II (CRS), or Scheme IV (CoC) via the official BIS Product Certification portal.

Step 2: Infrastructure & Testing Readiness

Establish internal testing laboratory facilities according to the BIS Scheme of Inspection and Testing (SIT) for the specific standard. Ensure all calibration certificates for testing equipment are valid and traceable to National Physical Laboratory (NPL) standards.

Step 3: Online Application Submission

Submit the formal application on the BIS Manak Online portal (www.manakonline.in). Upload required documents: premises proof, machinery list, test equipment calibration details, process flow charts, and test reports.

Step 4: Factory Audit & Sample Extraction

A BIS Inspecting Officer conducts an on-site audit of the manufacturing facility to verify process controls, testing competence, and extract independent samples for verification at a BIS Recognised Laboratory.

Step 5: Grant of Licence & Marking Implementation

Upon satisfactory audit results and passing lab test reports, BIS issues the Licence (CM/L number). The manufacturer must mark products strictly per BIS marking guidelines and pay required marking fees.

Step 6: Ongoing Post-Grant Surveillance

BIS maintains surveillance via unannounced factory audits and market sample collections. Non-conformity during surveillance testing may lead to licence suspension or cancellation under the BIS Conformity Assessment Regulations, 2018.

SECTION G & H — DIGITAL SERVICES & "KNOW YOUR STANDARD"

SERVICE_ID: SVC-001 | BIS Know Your Standard (KYS) System

Target Users:	Industries, Manufacturers, Researchers, Consumers
Purpose:	Allows search and discovery of published Indian Standards, draft standards, amendments, withdrawal status, and technical committee details.
Procedure:	1. Access KYS; 2. Search by IS Number or Keyword; 3. Filter by Technical Department; 4. View status, publication year, and download standard summary.
Official Portal:	BIS Know Your Standard Portal (Open Link)

SERVICE_ID: SVC-002 | BIS Care App & Licence Verification

Target Users:	Consumers, Retailers, Enforcement Officers
Purpose:	Verify genuineness of ISI-marked products, CRS registered electronics, and Gold Hallmarked jewelry (HUID).
Procedure:	Enter CM/L (Licence) number, CRS Registration Number, or 6-digit HUID code to instantly view status, manufacturer name, and validity.
Official Portal:	BIS Consumer Overview & Care Portal (Open Link)

SECTION J — GROUNDED RAG FAQ BASE

FAQ_ID: FAQ-001 | Standard vs. Quality Control Order

Question: What is the difference between an Indian Standard and a Quality Control Order (QCO)?

Verified Answer: An Indian Standard (IS) is a technical specification published by BIS defining safety, quality, or performance parameters. The existence of an Indian Standard is inherently voluntary for adoption. A Quality Control Order (QCO) is a statutory notification issued by a Ministry of the Government of India enforcing mandatory compliance with a specific Indian Standard. Once a QCO takes effect, no person can manufacture, import, store, or sell the specified product without a valid BIS licence and Standard Mark.

Source & Link: [BIS Product Certification FAQ \(Open Source\)](#)

FAQ_ID: FAQ-002 | Foreign Manufacturer Certification

Question: Can a foreign manufacturer apply for BIS certification for products under mandatory QCO?

Verified Answer: Yes. Foreign manufacturers exporting goods to India that fall under compulsory BIS certification must apply under the Foreign Manufacturers Certification Scheme (FMCS). The foreign manufacturer must establish a branch/ subsidiary in India or nominate an Authorized Indian Representative (AIR) residing in India to handle regulatory compliance and legal accountability.

Source & Link: [BIS FMCS Portal \(Open Source\)](#)

FAQ_ID: FAQ-003 | Consumer Complaints & Product Quality

Question: How can a consumer lodge a complaint regarding a sub-standard ISI-marked product?

Verified Answer: Consumers can lodge complaints regarding quality failures or misuse of the ISI/BIS mark through the "BIS Care" mobile application or via the official BIS online portal under the Consumer Complaints section. BIS conducts investigations, collects market samples, and may take administrative or legal proceedings against non-compliant manufacturers under the BIS Act, 2016.

Source & Link: [BIS Consumer Complaints Portal \(Open Source\)](#)

SECTIONS N, O & P — INTENT & GUARDRAILS DATASET

Intent Classification Reference

Intent Code	Description	Sample User Query
STANDARD_SEARCH	User searching for IS number by topic or product	"Find standard for solar panel"
MANDATORY_CERTIFICATION	User checking if certification is legally mandatory	"Is BIS mandatory for toys?"
QCO_SEARCH	User inquiring about Quality Control Orders/dates	"When did helmet QCO start?"
CERTIFICATION_PROCESS	User asking how to get a BIS licence / procedures	"How to apply for ISI mark?"
PRODUCT_VERIFICATION	User checking authenticity of CM/L, CRS, or HUID	"Verify CM/L 9512345678"
CONSUMER_COMPLAINT	User reporting sub-standard products or fraud	"Lodge complaint against ISI item"

Negative Guardrails & Refusal Rules

GUARDRAIL_ID: NEG-001 | Unspecific Product Mandatory Claims

Trigger Condition: User asks if certification is mandatory for an unspecific product (e.g., "Is steel mandatory?").

Correct System Action: Refuse direct Yes/No. Request precise sub-category, grade, or IS number.

Response Template: "BIS certification applicability depends on the specific product type, grade, and harmonized system (HS) code. For steel, multiple specific QCOs exist (e.g., steel wires, structural steel, rebar). Please provide the precise product designation or IS number to verify mandatory QCO status."

GUARDRAIL_ID: NEG-002 | Full Copyrighted Standard Request

Trigger Condition: User requests complete text, detailed drawings, or full engineering tables of a standard.

Correct System Action: Refuse reproduction of full text. Provide verified summary and link to official purchase page.

Response Template: "Full technical standards published by BIS are protected under copyright. I can provide the title, scope summary, amendments, and applicable QCO details. To access the complete text of Indian Standard [IS_NUMBER], please visit the official BIS Standards Portal: <https://standards.bis.gov.in/>"

SECTION Q & APPENDIX — SOURCE REGISTRY & RAG JSON

Official Source Registry

SRC-OFFICIAL-01 | Bureau of Indian Standards Main Portal

URL: <https://www.bis.gov.in/>

Type / Authority: Official Government Portal | Bureau of Indian Standards

Last Checked: 03 September 2026

SRC-OFFICIAL-02 | Products Under Compulsory Certification

URL: <https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

Type / Authority: Official Regulatory List | Bureau of Indian Standards

Last Checked: 03 September 2026

SRC-OFFICIAL-03 | BIS Manak Online E-Services Portal

URL: <https://www.manakonline.in/>

Type / Authority: Official Application & E-Services Portal | BIS

Last Checked: 03 September 2026

RAG JSON Direct Ingestion Payload

Below is the structured, self-contained JSON schema ready for vector database ingestion (e.g., Qdrant / Milvus / Pinecone):